

Course Code and Title: MAT110 Quantitative Methods
Lesson 3: Week 5-6
Topic: Correlation
Introduction Correlation analysis is a group of techniques to measure the association between two variables. One way of looking at the relationship between two variables is to portray the information in a scatter diagram. Correlation is a statistical measure that expresses the extent to which two variables are linearly related. It is a common tool for describing simple relationships without making a statement about cause and effect. The sample correlation coefficient, r, quantifies the strength of the relationship. Correlation cannot look at the presence or effect of other variables outside of the two being explored. Correlation also cannot accurately describe curvilinear relationships.
Correlation In many studies, we measure more than one variable for each individual. For example, we measure precipitation and plant growth, or number of young with nesting habitat, or soil erosion and volume of water. We collect pairs of data and instead of examining each variable separately (univariate data), we want to find ways to describe bivariate data, in which two variables are measured on each subject in our sample. Given such data, we begin by determining if there is a relationship between these two variables. As the values of one variable change, do we see corresponding changes in the other variable? We can describe the relationship between these two variables graphically and numerically. We begin by considering the concept of correlation. Correlation is defined as the statistical association between two variables. A correlation exists between two variables when one of them is related to the other in some way. A scatterplot is the best place to start. A scatter plot (or scatter diagram) is a graph of the paired (x, y) sample data with a horizontal x-axis and a vertical y-axis. Each individual (x, y) pair is plotted as a single point.

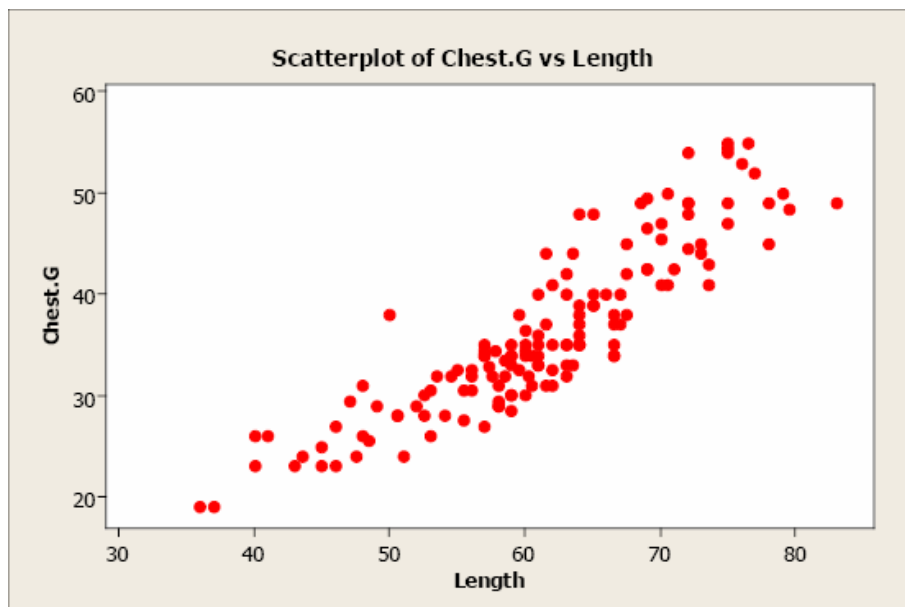


Figure 3.1: Scatter plot of chest girth versus length

In this example, we plot bear chest girth (y) against bear length (x). When examining a scatter plot, we should study the overall pattern of the plotted points. In this example, we see that the value for chest girth does tend to increase as the value of length increases. We can see an upward slope and a straight-line pattern in the plotted data points.

A scatterplot can identify several different types of relationships between two variables.

- A relationship has no correlation when the points on a scatter plot do not show any pattern.
- A relationship is nonlinear when the points on a scatterplot follow a pattern but not a straight line.
- A relationship is linear when the points on a scatterplot follow a somewhat straight-line pattern.

This is the relationship that we will examine.

Linear relationships can be either positive or negative. Positive relationships have points that incline upwards to the right. As x values increase, y values increase. As x values decrease, y values decrease. For example, when studying plants, height typically increases as diameter increases.

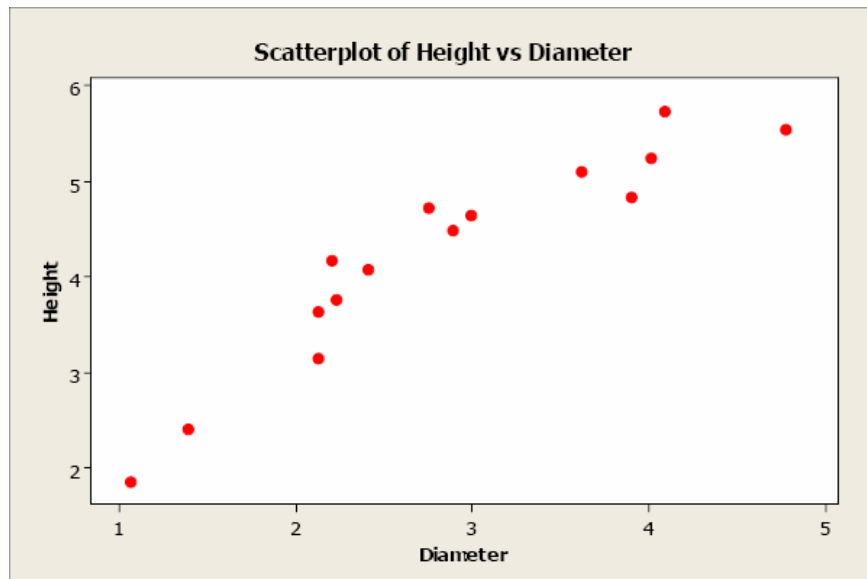


Figure 3.2: Scatter plot of height versus diameter

Negative relationships have points that decline downward to the right. As x values increase, y values decrease. As x values decrease, y values increase. For example, as wind speed increases, wind chill temperature decreases.

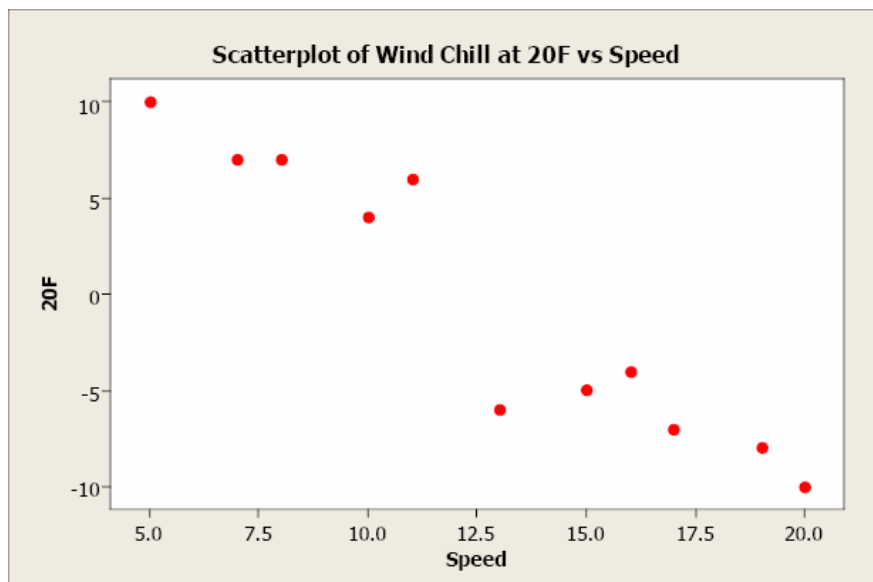


Figure 3.3: Scatter plot of temperature versus wind speed

Non-linear relationships have an apparent pattern, just not linear. For example, as age increases height increases up to a point then levels off after reaching a maximum height.

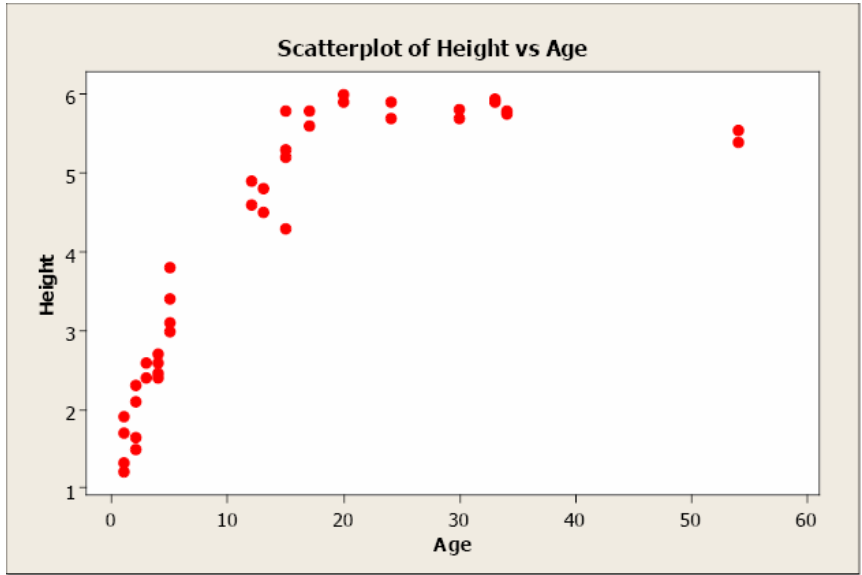


Figure 3.4: Scatter plot of height versus age

When two variables have no relationship, there is no straight-line relationship or non-linear relationship. When one variable changes, it does not influence the other variable.

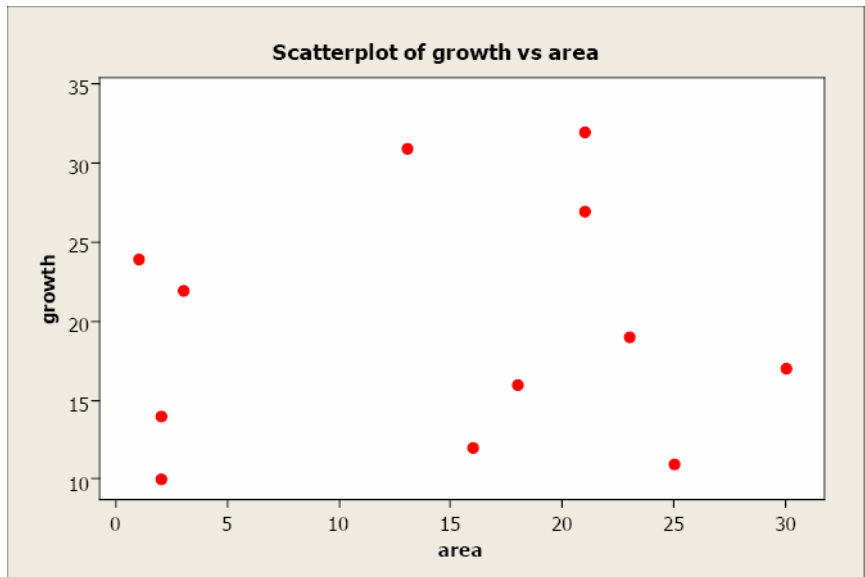


Figure 3.5: Scatter plot of growth versus area

Linear Correlation Coefficient

Correlation coefficients are used to measure how strong a relationship is between two variables. There are several types of correlation coefficient, but the most popular is Pearson's. Pearson's correlation (also called Pearson's R) is a correlation coefficient commonly used in linear regression.

Linear Correlation Coefficient Formula: Definition

Correlation coefficient formulas are used to find how strong a relationship is between data. The formulas return a value between -1 and 1, where:

- 1 indicates a strong positive relationship.
- -1 indicates a strong negative relationship.
- A result of zero indicates no relationship at all.

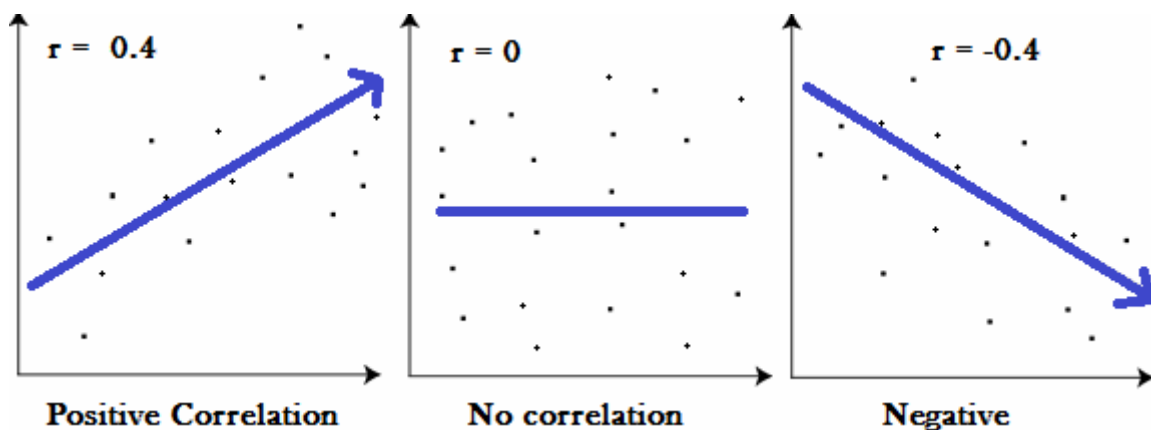


Figure 3.6: Graphs showing a correlation of -1, 0 and +1

Meaning

- A correlation coefficient of 1 means that for every positive increase in one variable, there is a positive increase of a fixed proportion in the other. For example, shoe sizes go up in (almost) perfect correlation with foot length.
- A correlation coefficient of -1 means that for every positive increase in one variable, there is a negative decrease of a fixed proportion in the other. For example, the amount of gas in a tank decreases in (almost) perfect correlation with speed.
- Zero means that for every increase, there is no positive or negative increase. The two are not

related.

The absolute value of the correlation coefficient gives us the relationship strength. The larger the number, the stronger the relationship. For example, $|-0.75| = 0.75$, which has a stronger relationship than 0.65 .

Examples of Positive Correlation

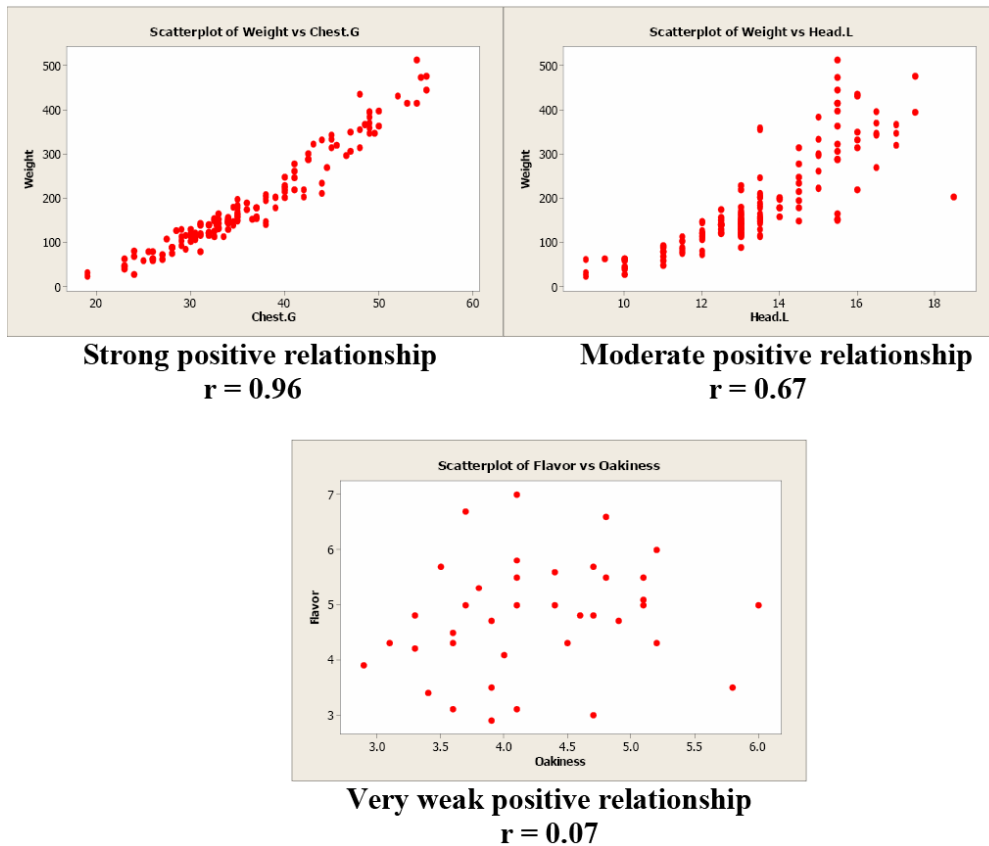
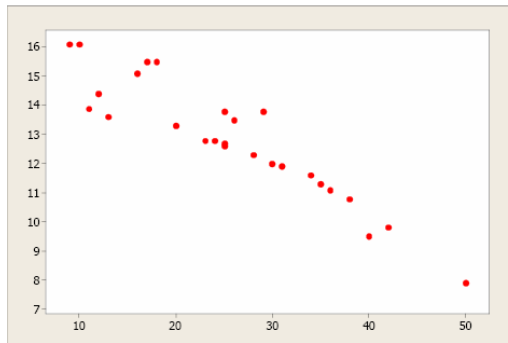
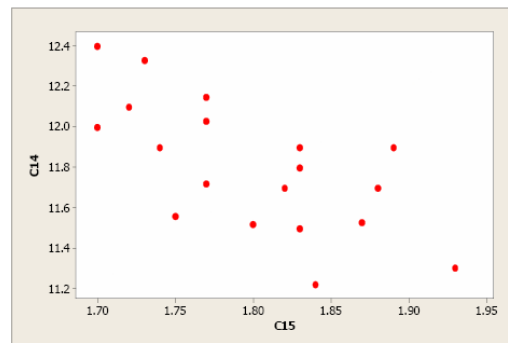


Figure 3.7: Examples of positive correlation

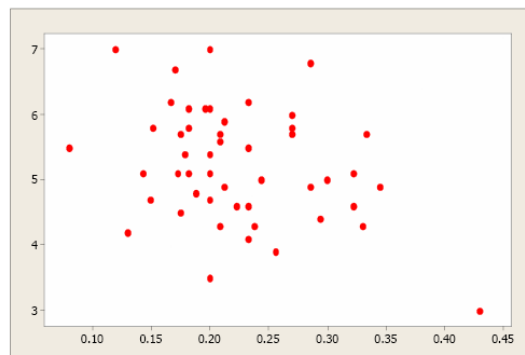
Examples of Negative Correlation



Very strong negative relationship
 $r = -0.93$



Moderately strong negative relationship
 $r = -0.67$



Very weak negative relationship
 $r = -0.13$

Figure 3.8: Examples of negative correlation

Note: Correlation is not causation! Just because two variables are correlated does not mean that one variable causes another variable to change.

Examine these next two scatterplots. Both of these data sets have an $r = 0.01$, but they are very different. Plot 1 shows little linear relationship between x and y variables. Plot 2 shows a strong nonlinear relationship. Pearson's linear correlation coefficient only measures the strength and direction of a linear relationship. Ignoring the scatterplot could result in a serious mistake when describing the relationship between two variables.

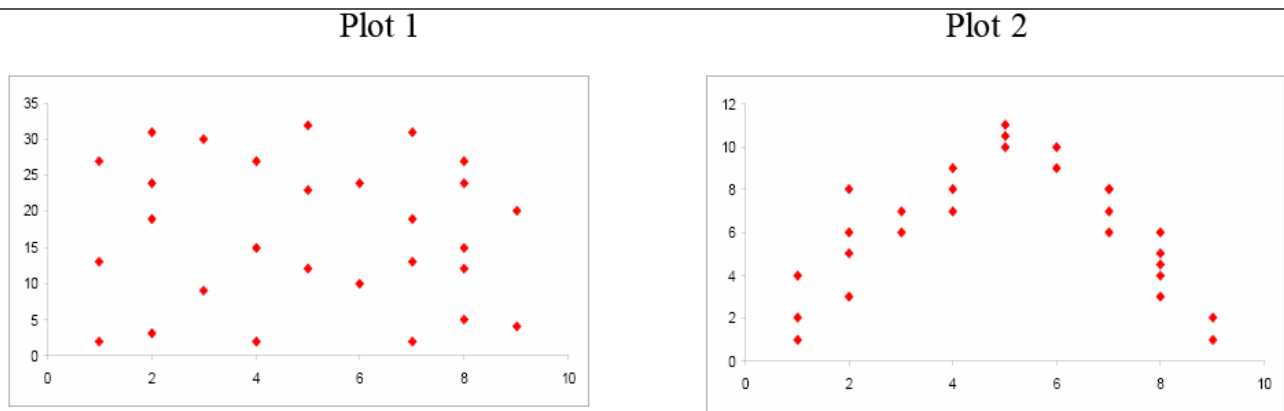


Figure 3.9: Comparison of scatterplots

When you investigate the relationship between two variables, always begin with a scatterplot. This graph allows you to look for patterns (both linear and nonlinear). The next step is to quantitatively describe the strength and direction of the linear relationship using “r”. Once you have established that a linear relationship exists, you can take the next step in model building.

Correlation Coefficient Formula

The Pearson’s correlation coefficient formula is

$$r = \frac{n(\sum xy) - (\sum x)(\sum y)}{\sqrt{[n\sum x^2 - (\sum x)^2][n\sum y^2 - (\sum y)^2]}}$$

The steps to calculate Pearson correlation coefficient are as follows.

1. Find out the number of pairs of variables, which is denoted by n. Example, let us presume x consists of 3 variables 6, 8, 10 and y consists of corresponding 3 variables 12, 10, 20.
2. List down the variables in two columns.

x	y
6	12
8	10
10	20

3. Find out the product of x and y in the 3rd column

x	y	x*y
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6	12	72
8	10	80
10	20	200

4. Find out the sum of values of all x variables and all y variables. Write the results at the bottom of the 1st and 2nd column. Write the sum of $x*y$ in the third column.

x	y	$x*y$
6	12	72
8	10	80
10	20	200
24	42	352

5. Find out x^2 and y^2 in the 4th and 5th columns and their sum at the bottom of the columns.

x	y	$x*y$	x^2	y^2
6	12	72	36	144
8	10	80	64	100
10	20	200	100	400
24	42	352	200	644

6. Insert the values in the formula then solve.

Example 1:

Sr No	Age (x)	Weight (y)
1	40	78
2	21	70
3	25	60
4	31	55
5	38	80
6	47	66

Solution:

Sr. No	Age (x)	Weight (y)	xy	x^2	y^2
1	40	78	3120	1600	6084
2	21	70	1470	441	4900
3	25	60	1500	625	3600
4	31	55	1705	961	3025
5	38	80	3040	1444	6400
6	47	66	3102	2209	4356
Total (Σ)	202	409	13937	7280	28365

The total number of people is 6, so $n = 6$

Find the Correlation Coefficient in Excel

<https://www.youtube.com/watch?v=ewK9ozyRh1g&t=7s>

What is Pearson Correlation?

Correlation between sets of data is a measure of how well they are related. The most common measure of correlation in statistics is the Pearson Correlation. It shows the linear relationship between

two sets of data. In simple terms, it answers the question, Can I draw a line graph to represent the data? Pearson correlation helps in knowing how strong the relationship between the two variables is. Not only the presence or the absence of the correlation between the two variables is indicated using the Pearson Correlation Coefficient, but it also determines the exact extent to which those variables are correlated. Using this method, one can ascertain the direction of correlation i.e., whether the correlation between two variables is negative or positive.

Potential problems with Pearson correlation

The Pearson correlation is not able to tell the difference between dependent variables and independent variables. For example, if you are trying to find the correlation between a high calorie diet and diabetes, you might find a high correlation of .8. However, you could also get the same result with the variables switched around. In other words, we could not say that diabetes causes a high calorie diet. Therefore, as a researcher you have to be aware of the data you are plugging in. In addition, the Pearson correlation will not give you any information about the slope of the line; it only tells you whether there is a relationship.

Real Life Example

Pearson correlation is used in thousands of real-life situations. For example, scientists in China wanted to know if there was a relationship between how weedy rice populations are different genetically. The goal was to find out the evolutionary potential of the rice. Pearson's correlation between the two groups was analyzed. It showed a positive Pearson correlation of between 0.783 and 0.895 for weedy rice populations. This figure is quite high, which suggested a fairly strong relationship.

Important Points

- The values can range from the value +1 to the value -1, where the +1 indicates the perfect positive relationship between the variables considered, the -1 indicates the perfect negative relationship between the variables considered, and a 0 value indicates that no relationship exists between the variables considered.
- It is independent of the unit of measurement of the variables. For example, if the unit of measurement of one variable is in years while the unit of measurement of the second variable is in kilograms, even then, the value of this coefficient does not change.

- The correlation coefficient between the variables is symmetric, which means that the value of the correlation coefficient between Y and X or X and Y will remain the same.

References:

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